UNIVERSITY OF MADRAS DEPARTMENT OF BIOCHEMISTRY	
Programme:	**M.Sc BIOCHEMISTRY**
Programme Code:	**LIFA**
Duration:	**2 years**
Programme Outcomes:	**PO1.** Acquire thorough knowledge and understanding of the basic concepts, theories and experimental methods to undertake a research career, either in an industry or in an academic setting. **PO2.** Develop problem solving skills, critical and analytical thinking. **PO3**. Develop a basic understanding of an interdisciplinary area and the ability to integrate across knowledge areas. **PO4**. Gain all the necessary practical skills required for biochemical analysis and bioinstrumentation. **PO5**.Effectively communicate disciplinary knowledge and scientific material clearly and correctly, in writing and orally. **PO6**. Function effectively as a leader or as a part of a team and develop skills in interdisciplinary teamwork. **PO7**.Commitment to good academic and work ethics and understanding of the ethical, safety and societal implications of practices relevant to the specific discipline. **PO8**. Recognize the need for continuous learning and updating within a discipline and provide the necessary skills to promote lifelong learning and career development. **PO9**. Know how to retrieve information from a variety of sources, including libraries, databases and the internet. **PO10.** Identifying, designing and executing a research problem, data analysis, interpretation, time and resource management.

Programme Specific Outcomes:	Programme Specific Outcomes (PSO)
	On successful completion of this course, students should be able to:
	PSO1. Understand the principles and methods of various techniques in Biochemistry, Biophysical chemistry, Molecular biology, Microbiology, Immunology, Enzyme kinetics and Cell biology. Based on their understanding of these techniques, they will have the ability to design and execute experiments during their final semester project and further research endeavors.
	PSO2. Acquire insight into the structure-function relationship of biomolecules, their synthesis and breakdown, the regulation of these pathways, and its importance in terms of clinical correlation. Students will also acquire knowledge of the principles of Nutritional biochemistry and apply them for the management of specific diseases.
	PSO3. Gain knowledge about the concepts of cellular signal transduction pathways and the association of aberrant signal processes to various diseases, acquire insight into the processes of membrane transport, protein sorting, folding and degradation, understand the mechanisms of immune responses and use this knowledge in the processes of immunization, vaccine development, transplantation and cancer immunotherapy.
	PSO4. Comprehend the central dogma of molecular biology, regulation of gene expression, molecular techniques used in rDNA technology, gene knock-out and knock-in techniques. Gain insights into morphogenesis and organogenesis, cell death mechanisms and regenerative therapy with stem cells and understand the relevance of gene therapy for the treatment of inherited diseases and cancer.
	PSO5. Be aware of good laboratory practices, learn how to review literature, learn the art of designing and executing experiments, be competent in analyzing, interpreting, citing available literature and presenting the data, learn to analyze the ethical and social responsibilities for conducting a scientific experiment and work as a part of a team.

List of Courses:

Semester	Course Code	Title of the Course	Core/Elective/ Soft Skill	Credits
I	1.LIF C001	Biochemical Techniques	Core	4
	2.LIF C002	Chemistry of Biomolecules	Core	4
	3.LIF C003	Clinical Biochemistry	Core	4
	4.LIF C004	Lab Course in Biochemical Techniques & Clinical Biochemistry	Core	4
	5.LIF E001	Human Physiology	Elective	3
	6.LIF E002	Nutritional Biochemistry	Elective	3
	UOM S001	Soft Skill I (Online)	S	2
II	6.LIF C005	Immunology	Core	4
	7.LIF C006	Intermediary Metabolism	Core	4
	8.LIF C007	Molecular Biology of the Gene	Core	4
	9.LIF C008	Lab Course in Molecular Biology	Core	4
	10.LIF E003	Enzymology	Elective	3
	11.LIF E004	Metabolic Regulation	Elective	3
	UOM S002	Soft Skill I (Online)	S	2
III	11.LIF C009	Hormonal Biochemistry	Core	4
	12.LIF C010	Industrial Microbiology	Core	4
	13.LIF C011	Lab Course in Microbiology	Core	4
	14.LIF C012	Eukaryotic Gene Expression	Core	4
	15.LIF E005	Biochemistry of Cell Signaling	Elective	3
	16.LIF E006	Developmental Biology	Elective	3
	UOM S003	Soft Skill 3 (Online)	S	2
	UOM I001	Internship	S	2
IV	17.LIF C013	Recombinant DNA Technology	Core	4
	18.LIF C014	Project and viva voce	Core	8
	19.LIF E007	Basics of Gene Therapy	Elective	3
	20.LIF E008	Molecular basis of diseases and therapeutic strategies	Elective	3
	UOM S004	Soft Skill 4 (Online)	S	2

Course	**LIFC001**
Title of the Course:	**BIOCHEMICAL TECHNIQUES**
Credits:	**4**
Pre-requisites, if any:	**Comprehensive Knowledge of Tools of Biochemistry**
Course Objectives	Biochemical techniques combine various inter-disciplinary methods in biological research and the course aims to provide students with the following objectives: 1. To understand the various techniques used in spectroscopy and centrifugation. 2. To explain chromatographic and electrophoretic techniques. 3. To comprehend the radiolabeling techniques and demonstrate knowledge of new microscopy techniques. 4. To acquire knowledge of mammalian cell culture techniques and aspects of plant tissue culture. 5. To demonstrate knowledge of various molecular biology techniques and uses of biosensors in biological research.
Course Outcomes	After completion of the course, the students should be able to: **CO1.** Attain good knowledge in modern spectroscopic and centrifugation techniques and apply the experimental protocols to plan and carry out simple investigations in biological research. (K1, K5) **CO2.** Demonstrate knowledge to implement the theoretical basis of chromatography and electrophoretic techniques in upcoming practical course work. (K3, K5) **CO3.** Tackle more advanced and specialized radioisotope and microscopic techniques that are pertinent to research. (K1, K2 & K5) **CO4.** Attain knowledge of molecular biology techniques and their applications in biological research. (K2, K3 & K5) **CO5.** Acquire thorough knowledge of theoretical and practical aspects of animal and plant cell culture techniques. (K1, K2 & K5)
Units	
I	Spectroscopy and Centrifugation: Regions of electromagnetic spectrum, Beer Lambert law, principle, instrumentation and applications of UV - visible spectrophotometry, Spectrofluorimetry - Assay of vitamins, NMR, ESR, Luminometry. Turbidimetry, Nephelometry, X - ray diffraction. Atomic absorption

	spectroscopy - principle and applications - Determination of trace elements. Basic principles of centrifugation. Preparative ultracentrifugation - Differential centrifugation, Density gradient centrifugation. Analytical ultracentrifugation - molecular weight determination.
II	Chromatography and Electrophoresis: Chromatography techniques - Paper, TLC - separation of lipids and amino acids, Gas liquid chromatography - Detector system, HPLC - Instrumentation - Pharmaceutical applications, Sepbox concept, Bioactivity guided fractionation, Ion exchange chromatography - measurement of glycosylated hemoglobin. Molecular exclusion chromatography, Affinity chromatography - principle, purification of mRNA. Electrophoresis of proteins - Native gels, SDS - PAGE, Isoelectric focusing, 2D PAGE, Detection, Estimation, and recovery of proteins in gels. Agarose gel electrophoresis of nucleic acids.
III	Radiolabeling and Microscopic Techniques: Radiolabeling techniques – Nature of radioactivity, decay and types of radiation. Radiation hazards and safety aspects. Radiation detection and measurements: Geiger-Muller counter, scintillation counter. Application of radioisotopes in biological science. Principles and applications of light, phase contrast microscopy. Principles of fluorescence, and confocal microscopy, Live-cell imaging, scanning and transmission electron microscopy, Scanning probe microscopy, Fixation and staining techniques. Histological techniques - Preparation of samples for histology.
IV	Cell culture techniques: Animal cell culture - laboratory design, cell culture equipments, aseptic techniques, culture vessels, culture media Trypsinization, Cryopreservation. Primary culture and cell line, organotypic culture, pluripotent stem cell lines, organ and embryo culture, culture of tumor cells, cytotoxicity, subculture.Safety, and bioethics. Plant tissue culture - media, micropropagation and somaclonal variation, protoplast culture, regeneration, and somatic hybridization.
V	Molecular Biology Techniques: Biochemical calculations, Purification and assay of proteins, DNA sequencing - enzymatic and chemical methods. Flow cytometry – Instrumentation and applications, Microarray, Blotting techniques - Western, Southern and Northern analysis. DNA finger printing, foot printing, PCR - principle and applications, RT - PCR, real time PCR. DNA and protein assays. Biosensors: principles, design, working, types and applications, electrochemical biosensors- potentiometric, amperometric biosensors, immunosensors, nanobiosensors.

Reading List (Print and Online)	Principles and techniques of biochemistry and molecular biology: https://www.kau.edu.sa/Files/0017514/Subjects/principals%20and%20techiniques%20of%20biochemistry%20and%20molecular%20biology%207th%20ed%
Recommended Texts	• Analytical Biochemistry: 3rd edition, **Holme DJ, Peck H**; Prentice Hall. • Biophysical chemistry, Principles and Techniques: 2016, **Upadhyay A, Upadhyay K and Nath N**; Himalaya Publishing House, India. • Principles and Techniques in Practical Biochemistry:7th edition, **Wilson K and Walker J**; Cambridge University Press. • Culture of Animal Cells: A manual of Basic techniques and Specialized Applications, 6th Edition, **Freshney I**; John Wiley Blackwell, Inc.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions.

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview.

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain.

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons.

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**S**	**L**	**M**	**S**	**S**	**L**	**L**	**S**	**S**	**M**
CO 2	**S**	**M**	**M**	**S**	**M**	**L**	**M**	**S**	**S**	**L**
CO 3	**S**	**M**	**L**	**S**	**M**	**M**	**M**	**S**	**M**	**L**
CO 4	**S**	**S**	**L**	**S**	**S**	**M**	**M**	**S**	**M**	**M**
CO 5	**S**	**S**	**M**	**S**	**M**	**M**	**M**	**S**	**M**	**M**

S-Strong M-Medium L-Low

Course	**LIF C002.**
Title of the Course:	**CHEMISTRY OF BIOMOLECULES**
Credits:	**4**
Pre-requisites, if any:	**Basic Knowledge of Biomolecules**
Course Objectives	The main objectives of this course are to: 1. Understand the structure of biomolecules. 2. Understand the significance of polysaccharides in biological processes. 3. Understand the concepts of protein structure and their significance in biological processes. 4. Know the structure, properties and biological significance of lipids in the biological system. 5. Understand the structures and functional roles of nucleic acids in the biological system.

Course Outcomes	**CO1.** Understand the chemical composition and 3-dimensional structure of biomolecules. (K1, K2) **CO2.** Learn about the structure of polysaccharides and their biological significance in living system. (K2, K3) **CO3**. Understand the structure and functions of proteins in the biological system. (K3, K5) **CO4.** Provide clear knowledge about lipids and their importance. (K2, K3) **CO5.** Recognize the structure, types, properties and functions of DNA and RNA. (K1, K2)
Units	
I	Structure of atom, Chemical composition and bonding, 3-dimensional structure, configuration and conformation, chemical reactivity. Water: Its effect on Biomolecules, Structure and properties of water, Ionization of water, weak acids and weak bases and their dissociation constants, pH, Buffers- definition, action, physiological buffers, pK, Henderson-Hasselbach equation.
II	Macromolecules and their monomeric subunits - 3 dimensional structures of proteins– peptide bonds, Ramachandran plot, alpha helix, parallel and anti-parallel beta pleated sheets. Bond angles and characteristic amino acid content. Tertiary and quaternary structures, Protein stability, denaturation, protein folding and chaperones. Structures of fibrous and globular proteins - myoglobin and hemoglobin - Bohr effect. Structure of collagen, keratin. Complementary interactions between proteins and ligands. Actin, myosin and molecular motors.
III	Sugars – Classification and reactions, Polysaccharides – properties and biological reactions, structure of starch, glycogen, cellulose and chitin, glycoconjugates – proteoglycans, glycoproteins and glycolipids. Homoglycans and Heteroglycans, Complex carbohydrates.
IV	Discovery of DNA, Structure of DNA, A form, B form of DNA, Z DNA, bent DNA, Biological significance of double stranded DNA, Topology of DNA - linking number, twist and writhe, DNA bending, supercoiling. Structure of different RNAs and their biological significance, Conformational properties of polynucleotides – secondary, tertiary structural features and analysis.
V	Lipids: Classification, structures and functions, Structure of storage lipids, structural lipids in membranes and their functions, glycerolipids, sphingolipids, phospholipids and eicosanoids (Thromboxanes, Leukotrienes and Prostaglandins).

Reading List (Print and Online)	Structural isomers and stereoisomers: https://www.youtube.com/watch?v=2AalKF41KC4 Essentials of Biomolecules: https://nptel.ac.in/courses/104/103/104103121/ DNA Structure: https://www.youtube.com/watch?v=EA6VV0WCNLE Definition and classification of Lipids https://www.youtube.com/watch?v=-3TmDPzyBeY
Recommended Texts	**1.** Biochemistry: 4th Edition, **Voet D and Voet JG**; John Wiley & Sons Inc **2.** Biochemistry: 6th edition, **Berg JM, Tymoczko JL, StryerL**; W.H.Freeman and Co. **3.** Lehninger Principles of Biochemistry, 7th edition**, Nelson DL, and Cox M**; WH Freeman and Co. **4.** Biochemistry: 4th edition, **Mathews CK and van Holde KE**; Benjamin/Cummings Publishing Co.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	s	L	M	S	M	M	L	S	M	M
CO 2	S	M	L	S	M	M	L	S	M	L
CO 3	S	M	L	S	S	L	L	S	M	L
CO 4	S	M	M	S	L	M	M	S	M	M
CO 5	S	S	M	S	S	M	M	S	M	M

S-Strong M-Medium L-Low

Course	**LIFC003**
Title of the Course:	**CLINICAL BIOCHEMISTRY**
Credits:	**4**
Pre-requisites, if any:	**Comprehensive knowledge of Basic Biochemistry**
Course Objectives	The candidates undertaking this course will understand the concept of normal and pathological ranges of biochemical components in our body. The following topics are covered. 1. Analysis of plasma proteins in health and disease. 2. Insight into the clinical aspects of liver disorders and liver function tests. 3. Insights into the clinical aspects of renal disorders and renal function tests. 4. Understand the importance of enzymes in the diagnosis of diseases. 5. Analysis of clinical parameters to understand normal ranges and abnormal ranges of biochemical components and hormones. An in-depth idea about hyperglycemia and inborn errors of metabolism.

Course Outcomes	**CO1.** Understand the role of plasma proteins in health and disease (K1, K2) **CO2.** Gain abundant knowledge about liver function tests, liver disorders and gastrointestinal diseases (K1, K3 & K5). **CO3.** Grasp the clinical significance of kidney function tests and associated disease settings (K1, K2 & K5). **CO4.** Good knowledge about clinical enzymes and principles of clinical lab tests (K1, K2). **CO5.** Learn the basics of inborn errors of metabolism, glycogen storage diseases and hyperglycemia (K1, K2 & K5)
Units	
I	Plasma proteins and their variation in diseases. Hemopoiesis and disorders of hemopoiesis, Hemoglobinopathies, anemias, hemorrhagic diseases. Normal and abnormal clotting mechanisms. Plasma lipids and lipoprotein changes in various diseases. Respiratory acidosis and alkalosis.
II	Clinical manifestations and biochemical changes in liver diseases (a) infectious–Viral hepatitis (b) toxic-alcohol-induced liver damage (c) genetic– hemochromatosis (d) immune – auto-immune hepatitis and biliary cirrhosis (e) neoplastic– hepatocellular carcinoma. Diagnosis of liver disorders with special reference to jaundice and cirrhosis. Liver function tests, Gastric function tests, Malabsorption syndrome, acidity, peptic ulcer, colon cancer.
III	Renal function tests - Diseases of the kidney (acute and chronic renal failure, diabetes insipidus, glomerulonephritis, nephrotic syndrome, uremic syndrome, renal hypertension, renal calculi, renal tubular acidosis). Drugs and toxins associated with renal diseases. Hemo dialysis and peritoneal dialysis.
IV	Clinical enzymology – Enzymes in plasma and their origin, general principles of assay, Clinical significance of enzymes and isoenzymes. (phosphatases, 5’ nucleotidase, gamma- glutamyltransferase, amylase, lipase, serum choline esterase, LDH, transaminases and creatine kinase), measurement of serum enzymes in diagnosis.
V	Clinical aspects of hyperglycemia- Diabetes mellitus, glucose tolerance test, oral hypoglycemic drugs, Inborn errors of metabolism- Glycogen storage diseases- VonGierke’ disease, Pompe’s disease, Anderson’s disease, Mcardle’s disease, Cori Forbes disease, Diseases

	related to amino acid catabolism- Tyrosinemia, Phenylketonuria, Maple syrup urine disease, Histidinemia, hyperprolinemia, Diseases associated with nucleotide metabolism-Gout, Lesch-Nyhan syndrome. Immunodeficiency disorders.
Reading List (Print and Online)	Introduction to clinical biochemistry : Dr.Graham Basten http://nsdl.niscair.res.in/bitstream/123456789/1089/1/introduction-to-clinical-biochemistry.pdf
Recommended Texts	• Textbook of Biochemistry with Clinical Correlations: 7th edition, **Devlin TM**; Wiley-Liss • Handbook of Clinical Biochemistry**:** 2nd edition, **Swaminathan R**; Oxford University Press, USA • Tietz Fundamentals of Clinical Chemistry: 6th edition, **Burtis CA and Ashwood ER**; Saunders Elsevier • Harper's Illustrated Biochemistry: 31st edition, • **Rodwell VW, Bender DA, Botham, KM, Kennelly PJ, Weil A**; McGraw-Hill

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	L	L	S	S	M	L	S	S	L
CO 2	S	L	M	S	S	M	M	S	S	M
CO 3	S	M	L	S	M	L	L	S	L	L
CO 4	S	M	L	S	S	M	M	S	M	M
CO 5	S	S	M	S	M	M	M	S	M	M

S-Strong M-Medium L-Low

Course	**LIFC004**
Title of the Course:	**LAB COURSE IN BIOCHEMICAL TECHNIQUES AND CLINICAL BIOCHEMISTRY**
Credits:	**4**
Pre-requisites, if any:	**Knowledge of Elementary Principles of Analytical Biochemistry (centrifugation, chromatography, colorimetric assays) and Metabolic Reactions**
Course Objectives	1. To instill skill in students enabling them to apprehend the wider knowledge about principles and techniques to be employed for the clinical analytes under investigation. 2. To inculcate good laboratory ethics in handling specimens and performing experiments. 3. To nurture good practices in learning and executing the techniques that are useful for research purposes. 4. To revise knowledge of enzymes and their nature by learning techniques of isolation, partial purification along with the activity assessment. 5. To achieve training in partition chromatographic technique. 6. To perform the isolation and identification of the organelles of a cell using differential centrifugation. 7. To perform colour reactions to diagnose the clinical condition of the biological samples reactions and interpret the observed data.

	8.(4A-C). To follow the experimental protocols in a step-wise manner, calculate and interpret the clinical inference and to learn the clinical significance. (With respect to: A: general assessment, B: liver function test, C: kidney function test.) • (4D). Cardiac function Test: to perform and to demonstrate the key clinical tests used to assess cardiac function. • (4E). To impart skills of drawing and handling blood samples for the assessment of blood components for an efficient interpretation of the overall health of an individual.
Course Outcomes	**CO1.**The student will be able to employ the relevant techniques for different laboratory tasks required for research. (K2, K4) **CO2.**The student will be able to collect, preserve, process and use appropriate techniques to examine the samples followed by a beneficial suggestion (K2, K4 & K6). **CO3.**The student will be fine-tuned in handling the clinical specimens and gain mastery in colorimetry, spectrophotometry and chromatographic techniques. (K4,K6) **CO4.**The student will be able to in diagnose the clinical condition of the samples provided and also can critically evaluate the same (4A-C). (K3,K4 & K6) **CO5.**The student, in addition to acquiring skill in performing cardiac tests, can also review the application of an electrocardiogram (4D).((K3,K4 & K6) **CO6.**The student can obtain skill in analyzing and diagnosing the clinical condition using blood cell counts and composition (4E) (K5, K6).
Units	
I	**1.** a) Purification of alkaline phosphatase from goat kidney. b) Determination of optimum pH and temperature of alkaline phosphatase. c) Determination of specific activity and Km of alkalinephosphatase. d) Effect of activators and inhibitors on the activity of alkaline phosphatase. d) Checking the purity using SDS-PAGE e) Separation of isoenzymes of Alkaline Phosphatase
II	**2.** TLC separation of lipids.

III	**3.** a) Fractionation of sub-cellular organelles by differential centrifugation. b) Assay of marker enzymes of the sub-cellular fractions
IV	4.Routine Clinical Assays A)General Procedures (i) Estimation of blood glucose (ii) Estimation of total Protein in serum B) Liver function tests (i) Metabolic function a) Estimation of Bilirubin b) Estimation of A/G ratio (ii) Clinical Enzymology a) Estimation of serum alanine transaminase (SGPT) b) Estimation of serum aspartate transaminase (SGOT) c) Estimation of serum lactate dehydrogenase d) Estimation of serum alkaline phosphatase e) Estimation of serum acid phosphatase C) Kidney function tests Group I tests Qualitative analysis of urine- General Principles Qualitative analysis of urine-Normal urine sample Qualitative analysis of urine-Pathological sample-I Qualitative analysis of urine-Pathological sample-II Group II Tests (i) Electrophoretic fractionation of serum proteins Group III tests (i) Estimation of Urea (ii) Estimation of Uric Acid (iii) Estimation of Creatinine (iv) Clearance Test Group IV tests (i) Estimation of Inorganic Phosphorus (ii) Estimation of Calcium D) Cardiac function Tests (i) Estimation of triacylgycerol (ii) Estimation of cholesterol (iii) Estimation of Creatine kinase (iv) ECG (Demo) E) Haematology a) Total count, differential count, erythrocyte sedimentation rate, hemoglobin, blood grouping. b) Clotting time, bleeding time, prothrombin time, PCV, MCH,

	MCHC, Platelet count
Reading List (Print and Online)	https://www.bioscience.com.pk/topics/pathology/clinical-pathology/item/823-test-for-detection-of-bilirubin-in-urine https://www.bioscience.com.pk/topics/pathology/clinical-pathology/item/824-test-for-detection-of-bile-salts-in-urine https://labmonk.com/determination-of-sgot-and-sgpt https://labmonk.com/colorimetric-analysis-of-bilirubin-in-serum
Recommended Texts	Textbook of medical laboratory technology, Clinical Laboratory Science and Molecular Diagnosis : 3rd Edition, **Godkar PB** and **Godkar DP ;** Bhalani Publishing House Practical Enzymology 3rd Edition, **Hans Bisswanger** ;Wiley

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Understand/Comprehend (K2)- MCQ, True/False, Concept explanations.

Application (K3)- Solve problems, Observe, Explain.

Analyse (K4)- Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas.

Evaluate (K5)- Critique or justify with pros and cons.

Create (K6)- Check knowledge in specific or offbeat situations, Discussion.

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	M	S	S	S	S	S	S	L	S
CO2	S	M	M	S	S	S	S	S	L	S
CO3	S	M	M	S	S	S	M	S	L	S
CO4	S	S	M	S	S	S	M	S	L	S
CO5	S	M	M	S	S	S	M	S	L	S
CO6	S	M	M	S	S	S	M	S	L	S

S-Strong M-Medium L-Low

Course	Elective, LIF E 001
Title of the Course:	**HUMAN PHYSIOLOGY**
Credits:	**3**
Pre-requisites, if any:	**The Students Ought to Exhibit Outlined Information on Structural and Functional Roles of the Vital Organs of the Human Body.**
Course Objectives	1. To employ the basic practical knowledge gained in clinical haematology, to apprehend the details of blood and its components in maintaining homeostasis. 2. To provide elaborate views and explain the concepts of neuronal functions and nerve muscle integration. 3. To enumerate the mechanistic aspects and regulatory modes of pulmonary and cardiac systems along with neurohormonal control. 4. To understand the formation of urine and regulation of renal function. 5. To dissect the mechanistic components of gastrointestinal tract and their role in achieving digestion and absorption.
Course Outcomes	**CO1.** The student will be able to delineate the structure of vital organs and their functional units that govern their mechanistic modalities in the maintenance of physiological balance of the body. (K1,K2 & K5) **CO2.** The student will be able to recognize the significance of blood in the maintenance of body homeostasis in addition to understanding the pathological issues. .(K1,K2 & K5) **CO3.** Will be able to explain and compare the neuro and neuromuscular components that govern the integrity of sensory/motor control. .(K1,K2 & K5) **CO4.** Will be able to apprehend and summarize the significant functional aspects of lung and heart which are involved in gaseous exchange and blood circulation in the body. .(K1,K2,K3 & K5) **CO5.** Will be able to clearly outline the structural and functional responsibilities of kidney that include excretion, acid base and electrolyte balance. .(K1,K2,K3 & K5) **CO6.** The students will assimilate and explain the details of mechanisms involved in digestion and absorption. .(K1,K2 & K5)

Units	
I	Composition of body fluids. Homeostasis. Plasma proteins and their functions. Formed elements, WBC and Platelets, Hemostasis and coagulation of blood, Mechanism of clotting – Clotting factors, clot retraction, fibrinolysis, disorders of clotting. CSF formation, composition and function.
II	Muscle &Nerve Muscle tissue – varieties – voluntary, involuntary, and cardiac- structure and functions. Nervous tissue – medullated and non-medullated. Structure of Neuron – Classification, functions, and properties of nerve fiber. Autonomous nervous system – Sympathetic and Parasympathetic functions, Neurotransmitters. Mechanism of conduction of nerve impulse. Synapse – Types, properties and functions, synaptic transmission. Functions of hypothalamus.
III	Chemistry of Respiration and Respiratory Control–Functional Anatomy of Respiratory tract, pulmonary ventilation. Mechanism of ventilation. Surfactant nature and function. Respiratory membrane and its importance. Transport of gases (O2, CO2). Introduction to circulation- Heart and valves of the heart- Origin of heart beat- Conduction through ventricular muscle- cardiac cycle-Metabolism and nutrition of heart muscle-Nerves of the heart and their action-Heart rate- Regulation of heart rate- Factors influencing cardiac output- Neurohormonal activation – calcium transport – contractile proteins.
IV	Renal Function: Structure and function of nephron, Renal blood flow and its importance, composition of urine, Functions of tubules, Nerve supply to urinary bladder, Micturition, Water, Electrolyte and Acid base balance.
V	Digestion and Absorption: Composition of saliva, Gastric and Pancreatic Secretions. Composition of Gastric and pancreatic juice, Factors affecting the secretions. Mechanism of secretion. Liver - Structure and function, secretions, composition and secretion of bile, gastrointestinal hormones and their actions, Absorption of carbohydrates, fats , proteins, vitamins, water and electrolytes.
Reading List (Print and Online)	An Overview of Blood | Anatomy and Physiology https://courses.lumenlearning.com/nemcc-ap/chapter/an- overview-of-blood/ Muscle and Nervous Tissues | Biology for Majors https://courses.lumenlearning.com/wm-biology2/chapter/muscle-and-nervous-tissues/ Human Biology (Wakim and Grewal)

	https://bio.libretexts.org/Bookshelves/Human_Biology/Book%3A_ The Kidneys and Osmoregulatory Organs https://opentextbc.ca/biology/chapter/22-2-the-kidneys-and-osmoregulatory-organs/ Digestive system regulation https://opentextbc.ca/biology/chapter/15-4-digestive-system-regulation/
Recommended Texts	Harper's Illustrated Biochemistry: 31st edition**, Rodwell VW , Bender DA, Botham, KM, Kennelly PJ, Weil A**; McGraw-Hill Textbook of Medical Physiology : 12th edition**, Guyton AC and Hall JE**; Saunders WB.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1)- Simple definitions, MCQ, Recall steps, Concept definitions

Understand/Comprehend (K2)- MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3)- Suggest idea/concept with examples, Explain

Evaluate (K5)-Longer essay/Evaluation essay, Critique or justify with pros and cons

Mapping course outcomes for each course with programme outcomes (PO)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**
CO2	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**
CO3	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**
CO4	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**
CO5	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**
CO6	**S**	**L**	**L**	**L**	**S**	**L**	**L**	**S**	**L**	**L**

S-Strong M-Medium L-Low

Course	**LIF E 002**
Title of the Course:	**NUTRITIONAL BIOCHEMISTRY**
Credits:	**3**
Pre-requisites, if any:	**Basic Knowledge of Biochemistry and Nutrition**
Course Objectives	**The major learning objectives of the course are:** 1. To introduce the various aspects of biochemistry related to macronutrients and micronutrients. 2. To provide information on the importance of nutrition under various physiological states. 3. To know the sources and nutritional importance of vitamins and minerals and list the disorders of their deficiencies. 4. To explore the consequences of nutrient deficiencies and to enlighten on dietary supplementation. 5. To understand the importance of dietary modifications and nutritional intervention in the management of specific diseases.
Course Outcomes	**On successful completion of this course, students should be able to:** **CO1.** know the importance of a balanced diet and apply biochemical and physiological principles to nutrition practice. (K1, K2 , K3, & K4) **CO2.** assess nutritional status of individuals in various life-cycle stages and translate nutrient needs into menus for individuals across the lifespan. (K1,K2,K3 & K4) **CO3.** know the sources, nutritional importance, diseases and manifestations associated with macro and micro nutrients (K1,K2 .K3 & K5) **CO4**. recognize special dietary considerations that need to be taken for individuals with certain medical conditions like, fever, obesity, and malnutrition. (K1, K2 , K3, K5 & K6) **CO5.** acquire knowledge and understanding of the principles of nutritional biochemistry and apply them for the management of specific diseases like atherosclerosis, ulcer, hypertension, diabetes and kidney diseases. (K1, K2 ,K3,K5 & K6)

Units	
I	Basic concepts - Nutrition - Concept - Composition of food - Macro and Micro nutrients and their functions. Measurement of the fuel values of foods. Direct indirect calorimetry. Basal metabolic rate: factors affecting BMR, measurement and calculation of BMR. Measurement of energy requirements. Specific dynamic action of proteins. National and regional food pattern in India.
II	Elements of nutrition - Plant and animal sources of simple and complex carbohydrates, fats and proteins and their requirement. Biological value of proteins. Concept of protein quality. Protein sparing action of carbohydrates and fats. Essential amino acids, essential fatty acids and their physiological functions. Role of fiber in diet.
III	Vitamins and Minerals- Dietary sources, biochemical function, requirements and deficiency diseases associated with vitamin B complex, Vitamin C and A, D, E & K vitamins. Nutritional significance of dietary calcium, phosphorus, magnesium, iron, iodine, zinc and copper. Nutritional requirement under different physiological states -infancy, childhood, pregnancy, lactation and ageing.
IV	Malnutrition - Diseases arising due to Protein - Calorie Malnutrition and under nutrition (Kwashiorkor and Marasmus), Prevention of malnutrition, Vitamin (fat and water soluble) deficiency diseases - Mineral deficiency diseases - symptoms and dietary supplementation.
V	Nutrition in diseases -Symptoms of diseases and modification of dietary pattern for patient suffering from fever (Typhoid and Malaria), Jaundice, hyper acidity (Ulcer), Atherosclerosis, Hypertension, kidney diseases and diabetes in adults. Starvation and Obesity.
Reading List (Print and Online)	Nutritional Biochemistry-Videos and Lessons. https://study.com/academy/topic/nutritional-biochemistry.html Nutritional and Clinical Biochemistry course https://onlinecourses.swayam2.ac.in/cec20_ag01/preview Biochemistry and Nutrition https://www.osmosis.org/library/foundational-sciences/biochemistry-and- nutrition

Recommended Texts	Advanced Text book on Food & Nutrition**:** 2nd Edition **;** **Swaminathan M,** Bappco publishers. Nutritional Biochemistry: 2nd edition, **Tom Brody;** Academic press. Normal and Therapeutic Nutrition: 17th edition**, Robinson C H, Lawler M R, CheiToweth W L and Garwick AE ;** Mac Millan publishing Co.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or offbeat situations, Discussion, Presentations

Mapping with Programme Outcomes:

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	**S**	**M**	**M**	**M**	**S**	**M**	**S**	**S**	**S**	**S**
CO 2	**S**	**S**	**M**	**L**	**S**	**M**	**S**	**S**	**S**	**S**
CO 3	**S**	**M**	**M**	**L**	**S**	**M**	**S**	**S**	**S**	**M**
CO 4	**S**	**S**	**S**	**M**	**S**	**M**	**S**	**S**	**S**	**S**
CO 5	**S**	**S**	**S**	**M**	**S**	**M**	**S**	**S**	**S**	**S**

S-Strong M-Medium L-Low

<table>
<tr><th>Course</th><th>LIFC005</th></tr>
<tr><td>Title of the Course:</td><td>IMMUNOLOGY</td></tr>
<tr><td>Credits:</td><td>4</td></tr>
<tr><td>Pre-requisites, if any:</td><td>Basic Knowledge of Immunology and Immune Disorders</td></tr>
<tr><td>Course Objectives</td><td>The candidates undertaking this course will understand the basics of immunology, immunological techniques and immune disorders. The following topics are covered.
1. Fundamentals of immunology and organization of immune system and its functions
2. Immunoglobulin structure and immunological techniques
3. Insights into major histocompatibility complex
4. Hypersensitivity reactions and Tumor immunology
5. An in-depth idea about immunization and common immune deficiency disorders</td></tr>
<tr><td>Course Outcomes</td><td>CO 1. Attain good knowledge in theories of ancient, modern and scientific immunology and to gain in depth knowledge about the organization of immune system. (K1, K2 & K3)
CO2. Summarize the immunological techniques used in biomedical research (K1, K2)
CO3. Gain in-depth knowledge of the concepts of immunity and major histocompatibility complex (K1, K2 & K3)
CO4. Gain insights into the hypersensitivity reactions (K1, K5)
CO5. Grasp the basics of immunization and immunization schedule and molecular basis of several immunological disorders (K1, K3 & K5)</td></tr>
<tr><td colspan="2">Units</td></tr>
<tr><td>I</td><td>Overview of the immune system: Background of Immunology & Historical perspectives, Innate and Adaptive immunity, Cells and tissues of the immune system – B and T lymphocytes, macrophages, Natural killer cells, Dendritic cells, properties of immunogenicity, Functions of lymphoid tissues, Hematopoiesis, Lymphoid organs – Bone marrow, Thymus, Lymphatic system, Myeloid system, Spleen. MALT. Stem cells. Immunity to bacteria and virus.</td></tr>
<tr><td>II</td><td>Antigen & Antibody:Immunoglobulin structure & functions.</td></tr>
</table>

	Immunoglobulin family, Antigen-antibody interactions, Antibody diversity, Measurement of antigen-antibody complexes, Complement system – Pathways, Functions and regulation. Diseases associated with the complement system. Essential features of antigens, Epitopes, paratopes, Cross reactivity, Essential factors for immunogenicity, Antigenic determinants, Immunological techniques – ELISA, RIA, Immunofluorescence microscopy, Western blot, in-situ localization methods – FISH, production and use of monoclonal antibodies.
III	Antigen processing and Histocompatibility antigens: Major histocompatibility complex, Organization of principal MHC genes in human (HLA) and mouse (H-2), functions of MHC, Antigen processing and presentation to T lymphocyte. Antigen receptors and accessory molecules of T lymphocytes. Activation of Lymphocytes. B cell activation and antibody production. Immune response- humoral and cell mediated immune response. T cell and B cell maturation, activation, proliferation, differentiation, regulation and effector functions.
IV	Hypersensitivity- Definition and types – factors governing hypersensitivity – Hypersensitive reactions. Congenital andacquired immunodeficiency. Immunological tolerance. Interleukins and Interferons- Production, biological functions and assay methods. Immunological tolerance. Transplantation: Bone marrow and organ transplants, graft versus host reaction and immunosuppression. Cancer Immunology: tumor antigens, immune response to tumor antigens, cancer immunotherapy and tumor vaccines.
V	Immunization and Immunodeficiency disorders Immunization: active and passive immunization. DNA based vaccines, Recombinant vaccines, peptide vaccines, Toxoid, Immunization schedule - BCG, small pox, DPT, Polio, measles, Hepatitis B. Adverse reactions from vaccines. Immunodeficiency: B and T – cell Immunodeficiency and Immunodeficiency diseases. Autoimmunity: Autoimmune diseases – AIDS, Rheumatoid arthritis, Systemic lupus erythematosus, Encephalomyelitis, Multiple sclerosis, Myasthenia gravis, Grave’s disease, Goodpasture syndrome, IDDM, Pernicious anemia.
Reading List (Print and Online)	Cellular and Molecular immunology https://nptel.ac.in/content/storage2/courses/102103038/download/module1.pdf

Recommended Texts	
	• Immunology: A Short Course: 7th Edition, **Coico R, Sunshine G**; Wiley- Blackwell • Immunology: 2nd edition , **Richard M Hyde**; Reston Pub. Co • Immunology: An Introduction: 4th edition. **Tizard I.R;** Saunders College Publishing • Cellular and Molecular Immunology: 9th edition, **Abbas AK , Lichtman A, Shiv pillai** ; Elsevier • Janeway's Immunobiology : 9th Edition, **Murphy KM and Weaver C**; Garland Science • Essentials of Immunology: 7th Edition, **Roitt IM**; Blackwell Scientific Publications.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**S**	**M**	**L**	**S**	**S**	**L**	**L**	**S**	**M**	**L**
CO 2	**M**	**M**	**L**	**S**	**S**	**M**	**M**	**S**	**M**	**M**
CO 3	**M**	**M**	**L**	**M**	**M**	**L**	**L**	**S**	**M**	**M**
CO 4	**S**	**L**	**M**	**M**	**S**	**L**	**S**	**S**	**M**	**M**
CO 5	**S**	**S**	**M**	**S**	**M**	**M**	**S**	**S**	**M**	**M**

S-Strong M-Medium L-Low

Course	**LIF C006**
Title of the Course:	**INTERMEDIARY METABOLISM**
Credits:	**4**
Pre-requisites, if any:	**Knowledge of the Underlying Biochemical Concepts and Elementary Ideas about Macromolecules**
Course Objectives	1. To impart knowledge of the basic principles that govern the various metabolic pathways and to enable the students to understand the interrelationship of carbohydrates, proteins and fat metabolism. 2. To understand the metabolic fate of carbohydrates 3. To explain basic steps of lipid metabolism and the energy factor involved. 4. To enlighten students about general pathways and inter conversion of amino acid metabolites 5. To help students understand the metabolism and regulation of nucleic acids.
Course Outcomes	**CO1.** The student will be able to recall, and understand free energy relationships/interaction of biochemical metabolism and will be able to understand energy balance/change (K1,K2, K3, K4 & K5) **CO2.** The student will be able to recognize the energetics, regulation of carbohydrate metabolism, and explain metabolic pathways of carbohydrates. (K1,K2, K3 & K5) **CO3.** The student will be able to identify and understand the metabolic fate of lipid molecules. (K1,K2 & K5) **CO4.** The student will be able to obtain lucid knowledge on amino acid and non-ribosomal protein biosynthesis. (K1,K2 & K5) **CO5.** The student will able to review and appreciate the pathways (metabolic and regulatory) of nucleotides and nucleic acid. (. (K1,K2 , K5)
Units	
I	Bioenergetics – Concept of free energy, standard free energy, determination of ΔG for a reaction. Relationship between equilibrium constant and standard free energy change, biological standard state &

	standard free energy change in coupled reactions. Biological oxidation-reduction reactions, redox potentials, relation between standard reduction potentials and free energy change (derivations and numericals included). High energy phosphate compounds – introduction, phosphate group transfer, free energy of hydrolysis of ATP and sugar phosphates along with reasons for high ΔG. Energy charge**.**
II	Carbohydrates – Glycolysis, energy balance sheet regulation of glycolytic sequence by enzymes and hormones ,citric acid cycle, its function in energy generation and biosynthesis of energy rich bond, pentose phosphate pathway and its regulation. Gluconeogenesis, glycogenesis and glycogenolysis, glyoxylate and Gamma aminobutyrate shunt pathways, Cori cycle, anaplerotic reactions, Entner-Doudoroff pathway, glucuronate pathway. Biosynthesis of lactose, sucrose and starch.
III	Lipids – Introduction, hydrolysis of tri-acylglycerols, α-, ß- and ω-oxidation of fatty acids. Oxidation of odd numbered fatty acids – fate of propionate, role of carnitine, degradation of complex lipids. Fatty acid biosynthesis, Acetyl CoA carboxylase, fatty acid synthase, ACP structure and function, Lipid biosynthesis, biosynthetic pathway for triacylglycerols, phosphoglycerides, sphingomyelin and prostaglandins. Metabolism of cholesterol and its regulation. Energetics of fatty acid cycle. Biosynthesis of Eicosanoids. Lipoprotein metabolism.
IV	Amino Acids – General reactions of amino acid metabolism - Transamination, decarboxylation, oxidative and non-oxidative deamination of amino acids. Special metabolism of methionine, histidine, phenylalanine, tyrosine, tryptophan, lysine, valine, leucine, isoleucine and polyamines. Urea cycle and its regulation. Oxidative degradation of amino acids : Proteolysis, Transamination, oxidative deamination, acetyl CoA, Alpha ketogutarate, acetoacetyl CoA, succinate, fumarate and oxaloacetate pathway, decarboxylation, urea cycle, Ammonia excretion, Peptides, polyamines, Porphyrins, gamma glutamyl cycle, glutathione biosynthesis, Non ribosomal Protein Biosynthesis.
V	Nucleotides – Biosynthesis and degradation of purine and pyrimidine nucleotides and their regulation. Purine salvage pathway. Role of ribonucleotide reductase. Biosynthesis of deoxyribonucleotides and polynucleotides including inhibitors of nucleic acid biosynthesis. Overview of metabolism of Porphyrins, Biosynthesis and degradation of heme

Reading List (Print and Online)	Amino Acid Metabolism: Introduction – Biochemistry | Lecturio YouTube Lecturio Medical Biochemistry of Carbohydrates and its Metabolism http://www.smarteach.com/course/biochemistry-of-carbohydrates-and-its-metabolism/ Purine & Pyrimidine Synthesis (de-novo) | Easy Biology Class https://www.easybiologyclass.com/nucleotide-biosynthesis-de-novo-salvage-purine-pyrimidine-nucleotides-cells/
Recommended Texts	• Biochemistry: 4th Edition , **Voet D and Voet JG;** John Wiley & Sons Inc • Lehninger Principles of Biochemistry, 7th edition, **Nelson DL, and Cox M;** WH Freeman • Biochemistry:4th edition, **Mathews CK and van Holde KE;** Benjamin/Cummings Publishing Co • Biochemistry: 6th edition, **Berg JM, Tymoczko JL, Stryer**

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1)- Simple definitions, MCQ, Recall steps, Concept definitions

Understand/Comprehend (K2)- MCQ, True/False, Short essays, Concept explanations, Short summary or overview
Application (K3)- Suggest idea/concept with examples, Explain

Analyse (K4)- Differentiate between various ideas.

Evaluate (K5)-Longer essay/Evaluation essay.

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	**S**	**M**	**S**	**M**	**S**	**M**	**M**	**S**	**S**	**M**
CO2	**S**	**M**	**S**	**M**	**S**	**M**	**M**	**S**	**S**	**M**
CO3	**S**	**L**	**S**	**L**	**S**	**L**	**L**	**S**	**S**	**L**
CO4	**S**	**L**	**S**	**L**	**S**	**L**	**L**	**S**	**S**	**L**
CO5	**S**	**M**	**S**	**M**	**S**	**M**	**M**	**S**	**S**	**M**

S-Strong M-Medium L-Low

Course	**LIFC007**
Title of the Course:	**MOLECULAR BIOLOGY OF THE GENE**
Credits:	**4**
Pre-requisites, if any:	**Knowledge of the Basics of Cell and Molecular Biology**
Course Objectives	Molecular biology of the gene comprises of the structure and function of genes in the context of understanding the mechanics of replication, repair, transcription and translation processes in prokaryotes. The aim of this course is to understand the various levels of gene regulation coined under the following objectives. 1.To understand the process of replication in molecular detail and various repair mechanisms 2.To summarize the concepts of transcription and translation processes in prokaryotic system 3.To rationalize the concepts of gene regulation 4.To understand the models of recombination 5.To illustrate different concepts and aspects of Cell cycle regulation and molecular biology of cancer.
Course Outcomes	**CO1.**Understand how the DNA in a genome is organized, replicated, and repaired. The student will be excited to understand the mechanics of replication and various DNA repair processes (K1,K2) **CO2.**Demonstrate knowledge on prokaryotic gene expression. The course will excite the student to understand basic science and current concepts of gene expression (K1,K2,K3)

	CO3.Be able to give examples of and describe different types of mechanisms of gene regulation at transcriptional and posttranscriptional levels. Solve problems related to gene regulation. (K1,K4,K5) **CO4.**Be able to interpret the models of recombination (K1,K5) **CO5**.Understand the regulation of cell cycle and the molecular biology of cancer. (K1,K3,K5)
	Units
I	DNA replication and repair: Enzymes of replication, prokaryotic replication mechanisms, the cellular replisome, eukaryotic DNA replication, the role of topoisomerases and telomerase, regulation of replication. DNA repair mechanisms – Direct repair, excision repair, mismatch repair, recombination repair, SOS response, eukaryotic repair systems.
II	Prokaryotic gene expression**:** Transcription – subunits of RNA polymerase, □factor and promoter recognition, alternative □ factors, *E. coli* promoters, Rho-dependent and independent termination of transcription. Translation – organization of the ribosome, the genetic code, its universality, deciphering the genetic code, wobble hypothesis, activation, initiation, elongation and termination of translation in *E. coli*. The role of tRNA and rRNA, suppressor tRNAs and inhibitors of protein synthesis
III	Regulation of prokaryotic gene expression**:** The *lac* operon, identification of operator and regulator sequences by mutations, induction and repression, Foot-printing and gel-shift assays for identification of protein-DNA interactions. Catabolite repression. Trp operon – Attenuation, alternative secondary structures of trp mRNA. Regulation of translation by small RNA molecules, Control of lytic development in □phage.
IV	Recombination and mobile genetic elements**:** The Holliday model, general recombination in *E.coli*, site-specific recombination, transposons, transposition, phase variation in *Salmonella*, Retroviruses and retroposons.

V	Cell cycle and growth regulation**:** Cyclins, cyclin – dependent kinases, molecular basis of cell cycle regulation, role of p53 and Rb proteins, checkpoints in cell cycle regulation, Apoptosis – various stimuli and pathways. Oncogenes and cancer**:** Molecular Mechanisms-Proto-oncogenes, mechanisms of activation of proto-oncogenes, regulation of gene expression by oncoproteins, viral oncogene products, growth factor receptor kinases, components of signal transduction cascade.
Reading List (Print and Online)	Molecular Biology https://onlinecourses.swayam2.ac.in/cec20_ma13/preview Molecular Biology https://mooc.es/course/molecular-biology/ Molecular Biology Free Online Course by MIT Pat 3: RNA Uploaded by edX
Recommended Texts	• Lewin's Genes XII **:** 12th edition**, Krebs JE, Goldstein ES, Kilpatrick ST** ; Prentice Hall • Molecular Biology of the Gene : 6th edition**, Watson JD , Baker TA, Bell S, Gann A, Levine M, Losick R;** Cold Spring Harbor Laboratory Press • Essential Cell Biology **:**3rd edition, **Alberts B, Bray D, Hopkin K, Johnson A, Lewis J, Raff M, Roberts K, Walter P ;** Garland Science • Molecular Cell Biology : 8th edition **, Lodish H, Arnold Berk**; W.H.Freeman & Co Ltd

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:
Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	S	S	M	S	S	M
CO 2	S	L	M	M	S	M	M	S	M	M
CO 3	S	M	S	M	L	L	L	S	M	L
CO 4	S	S	S	S	S	S	M	S	M	L
CO 5	S	S	M	S	L	M	M	S	M	M

S-Strong M-Medium L-Low

Course	LIF C008
Title of the Course:	LAB COURSE IN MOLECULAR BIOLOGY
Credits:	4
Pre-requisites, if any:	**The students are Expected to Apply Theoretical Concepts in Molecular Biology for Effective Acquirement of Skills.**
Course Objectives	1.To isolate the chromosomal and extrachromosomal DNA. 2.To detect the isolated DNA using electrophoresis. 3.To prepare the cells to accept the foreign DNA and to detect the insertion. 4.To identify the expression of DNA qualitatively and quantitatively. 5.To isolate RNA and to amplify the molecule. 6.To learn and employ the knowledge of softwares and databases in analysing the sequences under investigation.
Course Outcomes	**CO1.**The candidate will be able to acquire knowledge and skill in techniques that are widely employed in research (K2,K3,K4) **CO2.**The candidate will be able to differentiate circular and linear DNA and detect them using their properties (K1, K2, K3, K4). **CO3.** Be able to prepare and identify the successful clones (K1, K2, K3, K4). **CO4**.Will be employing the theoretical knowledge in the detection of expression of genes in normal as well as pathological samples (K1, K2, K3, K4, K5). **CO5.** Will know the stringent steps of isolating and detecting RNA from the biological samples (K2,K3,K4) . **CO6.**Able to learn the usage of software tools for nucleic acid research (K3).
Units	
I	Isolation of plasmid DNA and Genomic DNA, Restriction enzyme cleavage of chromosomal DNA and plasmid DNA
II	Agarose gel electrophoresis of DNA
III	Preparation of competent *E. coli* cells, Transformation of plasmid DNA.

	SDS-PAGE, Western blotting, Southern blotting ,ELISA
IV	PCR, RT-PCR
V	Basic BLAST search, Sequence analysis using computer software
Reading List (Print and Online)	https://www.ncbi.nlm.nih.gov/probe/docs/techpcr/ http://universe84a.com/collection/gel-electrophoresis/ https://www.ncbi.nlm.nih.gov/
Recommended Texts	Molecular Biology Techniques, A classroom laboratory manual: 4th Edition, **Carson S, Miller H, Srougi M, Witherow DS**; Elsevier

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1)- Simple definitions, Recall steps, Concept definitions

Understand/Comprehend (K2)- MCQ, True/False, Concept explanations.

Application (K3)- Observe, Explain, solve problems

Analyse (K4)- Finish a procedure in many steps, Differentiate between various ideas.

Evaluate (K5)- Justify

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	S	S	S	S	S	M	S	M	M	M
CO2	S	S	S	S	S	M	S	M	M	S
CO3	S	S	S	S	S	M	S	M	M	S
CO4	S	S	S	S	S	M	S	M	M	S
CO5	S	S	S	S	S	M	S	M	M	S
CO6	S	S	S	L	L	M	M	S	S	S

S-Strong M-Medium L-Low

Course	LIF E 003
Title of the Course:	**ENZYMOLOGY**
Credits:	**3**
Pre-requisites, if any:	**Basic Knowledge about Kinetics and Chemical Reaction Mechanisms**
Course Objectives	**The major learning objectives of the course are :** 1. To provide a basic understanding of enzymes and introduce students to the theory and practice of enzymology. 2. To understand the kinetics of enzyme catalyzed reactions and determine the kinetic constants in the absence and presence of inhibitors. 3. To describe the role of regulatory enzymes in controlling metabolic pathways and cellular responses. 4. To explain the various mechanisms of enzyme catalysis and the factors that affect catalysis. 5. To define and discuss the biological role of isoenzymes and the use of enzymes in clinical diagnosis. 6. To comprehend the methods of immobilization of enzymes and realize their industrial applications.
Course Outcomes	**On successful completion of this course, students should be able to:** **CO1**.Isolate and characterize enzymes. (K1,K2 & K3) **CO2**.Analyze enzyme kinetic data, calculate kinetic parameters, and determine the mechanism of inhibition by a drug/chemical. (K1,K2 , K3 &K4) **CO3**.Differentiate Michaelis-Menten kinetics from sigmoidal kinetics, explain allosterism and the control of metabolism through enzyme regulation. (K1,K2 , K3 &K4) **CO4**.Describe the catalytic mechanisms employed by enzymes, in general, and specifically, serine proteases. (K1,K2 , K3 & K5) **CO5**.Analyze options for applying enzymes and their inhibitors in medicine and industries. (K1,K2 & K3)

Units	
I	Enzymes: basic definition, specificity of enzyme action-group specificity, absolute specificity, substrate specificity, stereochemical specificity, Nomenclature and classification of enzymes.Active site, Lock and key hypothesis, Induced fit hypothesis, Identification of amino acids at the active site-trapping of ES complex, identification using chemical modification, and site-directed mutagenesis.Measurement of enzyme activity, Specific activity. Criteria for purity of an enzyme
II	Enzyme Kinetics-Thermodynamics of enzyme action, Activation energy, transition-state theory. Single substrate enzyme catalyzed reactions -assumptions, Michaelis-Menten and Briggs-Haldane kinetics, derivation of equations. Double reciprocal (Lineweaver-Burk) and single reciprocal (Eadie -Hofstee, Hanes) linear plots , their advantages and limitations. Analysis of kinetic data- determination of Km, kcat, catalytic efficiency, Enzyme inhibition: Competitive, non-competitive and uncompetitive inhibition, Graphical analysis (Primary and secondary kinetic plots), Michaelis -Menten equation in the presence of these inhibitors. Determinations of kinetic constants in the presence of inhibitors. Suicide and feedback inhibition
III	Allosteric enzymes: Cooperativity, MWC and KNF models of allosteric enzymes. Sigmoidal kinetics taking ATCase as an example. Regulation of enzyme activity, Regulatory enzymes in carbohydrate metabolism (glycolysis, TCA cycle) Bi - Substrate reactions: Single Displacement reactions (Ordered and Random bi bi mechanisms), Double Displacement reactions (Ping pong mechanism), Examples, Graphical analysis, Determination of kinetic constants starting from Alberty’s equation.
IV	Mechanisms of enzyme catalysis: acid-base catalysis, covalent catalysis, electrostatic catalysis, metal ion catalysis, proximity and orientation effects. Mechanism of action of chymotrypsin and carboxypeptidase. Factors affecting catalysis - pH, temperature, enzyme and substrate concentration.
V	Isoenzymes taking LDH as an example. Immobilized enzymes: Techniques of immobilization, Characteristics of immobilized enzymes. Applications in industry and medicine. Enzymes as diagnostic reagents.

Reading List (Print and Online)	Enzymes | MIT OpenCourseWare | Free Online Course Materials https://ocw.mit.edu/high-school/biology/exam-prep/chemistry-of-life/enzymes/ Enzymology https://onlinecourses.swayam2.ac.in/cec20_bt20/preview https://mooc.es/course/enzymology/ The active site of enzymes https://dth.ac.in/medical/courses/biochemistry/block-1/1/index.php
Recommended Texts	• Enzymes: Biochemistry, Biotechnology and Clinical chemistry, 2nd edition**, Palmer T and Bonner P**; Affiliated- East West press private Ltd, New Delhi • Fundamentals of Enzymology, 3rd edition, **Price NC and Stevens L;** Oxford University Press. • Biochemistry:4th Edition**, Voet D and Voet JG;** John Wiley & Sons Inc • Lehninger Principles of Biochemistry, 7th edition**, David. L.Nelson, and Michael M. Cox;** WH Freeman

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps

Evaluate (K5) - Longer essay/ Evaluation essay

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	S	S	S	S	M	M	S	S	M
CO 2	S	S	S	S	S	M	M	S	M	S
CO 3	S	S	S	S	S	M	M	S	M	M
CO 4	S	S	S	S	S	L	L	S	M	M
CO 5	S	S	S	S	S	S	S	S	S	S

S-Strong M-Medium L-Low

Course	**LIF E004**
Title of the Course:	**METABOLIC REGULATION**
Credits:	**3**
Pre-requisites, if any:	**Basic Knowledge of Biochemistry**
Course Objectives	The main objectives of this course are : 1.To know the importance of metabolic flux and its regulation mechanisms. 2.To learn carbohydrate, amino acid metabolism and their regulation mechanisms. 3.To understand fatty acid synthesis and degradation and their regulation mechanisms. 4.To study the interrelationship of carbohydrate, lipid, protein and nucleic acid metabolism

Course Outcomes	**CO1**.Gain knowledge of the importance of metabolic flux and its regulation mechanisms with associated disorders. (K1 , K2) **CO2**.Understand carbohydrate metabolism and its regulation process by hormones and other factors.(K3) **CO3**.Able to understand the metabolic pathways of amino acids and their regulation. (K2 , K5) **CO4**. Understand the processes of fatty acid synthesis, degradation and their regulation. (K3 , K5) **CO5**.Understand the metabolism of nucleic acids and their regulation (K5)
	Units
I	Metabolic flux: Control of metabolic flux, flux generation, rate determining step in the control of pathway flux, Allosteric control, covalent modification, substrate cycles. Regulation of Glycolysis, role of PFK in Glycolysis, Role of Fructose 2,6 bis phosphate in muscle and Liver, regulation of gluconeogenesis,.Role of inhibitors. Regulation of Pyruvate dehydrogenase
II	Regulation of Glycogenesis, cAMP dependent protein kinases. Phosphorylase activation and inactivation. Co-ordinated control of Glycogenesis and Glycogenolysis. regulation, of HMP shunt, TCA cycle and Glyoxylate cycle regulation.
III	Electron Transport Chain: Mitochondrial structure and enzyme distribution. Electron carriers, their redox potential, their sequence, Electron transport and production of ATP. oxidative phosphorylation, regulation, inhibition of oxidative phosphorylation, chemiosmotic theory, mechanism of ATP synthase reaction, transport of substrates in and out of the mitochondria, uncouplers.
IV	Regulation of fatty acid metabolism, Hormonal Regulation, Sites of regulation. Control of cholesterol biosynthesis and transport. LDL-receptor activity and cholesterol homeostasis.
V	Regulation of Urea cycle, purine and pyrimidine metabolism . Regulation of heme biosynthesis in Erythroid and hepatic tissue. Integration of metabolism.

Reading List (Print and Online)	Metabolic reactions in biological system: https://www.youtube.com/watch?v=Lh-OBavYzM0 Introduction to electron transport chain: https://www.youtube.com/watch?v=COMD8yHJtLI Lipid metabolism: https://dth.ac.in/medical/courses/biochemistry/block-5/4/index.php Biochemical engineering: https://nptel.ac.in/courses/103/105/103105054/
Recommended Texts	• Biochemistry: 4th Edition ,**Voet D and Voet JG;** John Wiley & Sons Inc • Lehninger Principles of Biochemistry, 7th edition, **Nelson DL, and Cox M;** WH Freeman • Biochemistry:4th edition, **Mathews CK and van Holde KE;** Benjamin/Cummings Publishing Co • Biochemistry: 6th edition, **Berg JM, Tymoczko JL, Stryer L**; W.H. Freeman

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or offbeat situations, Discussion, Debating or Presentations

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	M	M	L	S	M	M
CO 2	S	S	M	S	M	S	M	S	M	M
CO 3	S	S	S	S	S	L	M	S	L	L
CO 4	S	M	L	S	L	M	L	S	L	L
CO 5	S	M	M	S	S	M	M	S	M	L

S-Strong M-Medium L-Low

Course	**LIF C009**
Title of the Course:	**HORMONAL BIOCHEMISTRY**
Credits:	**4**
Pre-requisites, if any:	**Basic knowledge of Hormones and Their Action**
Course Objectives	1.To explain to the students about hormones , their classification and biosynthesis 2.To explain about the mechanisms of hormone action and feedback mechanisms 3.To describe the hormones of hypothalamus and pituitary 4.To describe about the pancreatic hormones and their action 5.To describe about thyroid, parathyroid , adrenal and gonadal hormones
Course Outcomes	**CO1**.A student will know about hormones and their classification, biosynthesis and circulatory levels of each hormone (K1, K2 & K3) **CO2.** Understand the mechanisms of hormone action and feedback mechanisms (K1, K2 & K3)

	CO3.Understand the hormones of hypothalamus and pituitary (K1, K2 & K3) **CO4**.Learn about the pancreatic hormones and their action (K1, K2 & K3) **CO5**.Learn about the thyroid, parathyroid , adrenal and gonadal hormones (K1, K2 & K3)
Units	
I	Hormones – Classification, Biosynthesis, circulation in blood, modification and degradation. Mechanism of hormone action, Target cell concept – Feedback control and regulation. Hormones of Hypothalamus and pituitary – Vasopressin and oxytocin, Hypothalamic releasing factors. Anterior pituitary hormones – actions and feedback regulation of synthesis. Growth promoting, Lactogenic hormones. Glycoprotein hormones, the POMC family, Endorphins
II	Pancreatichormones – cell types of the islets of Langerhans. Insulin – structure, Biosynthesis, regulation of secretion, Biological actions and mechanism of action. Glucagon, somatostatin and pancreatic polypeptide. Insulin like growth factors – structure, biological action. Gastrointestinal hormones – secretin, gastrin, cholecystokinin – biological action, regulation of secretion.
III	Thyroid hormones – synthesis, secretion, transport, biological action, metabolic fate and mechanism of action, regulation. Parathyroid hormone – biological action, regulation of calcium and phosphorus metabolism and the role of calcitonin. Calcitriol – Biosynthesis, transport, functions, mechanism of action.
IV	Adrenal hormones – Glucocorticoids, mineralocorticoids, synthesis, secretion, transport, metabolism and excretion. Biological effects. Mechanisms of action, adrenal androgens, metabolic effects and functions. Adrenal medulla – Catecholamines, biosynthesis, storage, metabolism, regulate of synthesis. Chemical nature and biological action of prostaglandlins.
V	Gonadal Hormones – Chemical Nature. Biosynthesis, metabolism and metabolism of action of androgen, estrogen and progesterone. Factors involved in the regulation of gonadal hormone activities. Ovarian cycle. Pregnancy, biochemical changes in pregnancy.

Reading List (Print and Online)	Hormone Biochemistry https://www.youtube.com/watch?v=MHOpVy8VcXk Hormone Mechanism of Action https://www.youtube.com/watch?v=w2tp8VIkbIU Endocrine system physiology | human endocrine hormones https://www.youtube.com/watch?v=T6mPS53Jb1M
Recommended Texts	• Harper's Illustrated Biochemistry:31st edition,**Rodwell VW , Bender DA, Botham, KM, Kennelly PJ, Weil A;** McGraw-Hill • Clinical Endocrinology :2012,**Whitehead S and MiellJ** ; Scion Publishing • Textbook of Endocrinology : 14th edition**, Williams RKH ; Elsevier** • Basic and Clinical Endocrinology : 3rd edition, **Greenspan FS, Gardner DG;** McGraw Hill

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:
Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	S	M	L	M	M	M	S	M	L
CO 2	S	M	S	L	M	M	M	S	S	L
CO 3	M	M	S	M	M	L	M	S	M	M
CO 4	M	L	M	L	L	L	M	S	M	M
CO 5	S	L	S	M	L	M	M	S	M	L

S-Strong M-Medium L-Low

Course	**LIF C010**
Title of the Course:	**INDUSTRIAL MICROBIOLOGY**
Credits:	**4**
Pre-requisites, if any:	**Basic Knowledge of Microbiology**
Course Objectives	1. To gain knowledge of the structure, classification and use of microorganisms in various industries. 2. To know various fermenter designs, culture systems and the application of fermentation process in industry. 3. To understand the production and purification of fermented products and their industrial applications. 4. Understand the basic concepts of food and agricultural microbiology.

Course Outcomes	**CO1**.Students will be able to understand the structure and classification of microorganisms (K2 , K4) **CO2**.Gain knowledge of the uses of microorganisms in various industrial applications (K3 , K4) **CO3**.Understand the concepts of fermentation process, harvest and recovery. (K1 , K5) **CO4**.Students will know the types of microbial fermentation processes and their applications in pharmaceutical industry. (K2 , K3) **CO5**.Students will learn about the use of microorganisms in beverages, diary and food industries. (K3 , K6)
Units	
I	Structure of bacteria, fungi and viruses and their classification. Types and characteristics of microorganisms used in Industry (a) Food Industry (b) Chemical Industry (c) Pharmaceutical Industry
II	Fundamentals and principles of microbial fermentation techniques – application in industry and pharmaceutical Biochemistry. Fermentation – types, techniques, design and operation of fermenters including addition of medium. Types and characteristics of microorganisms, environmental conditions required for the growth and metabolism of industrially and pharmaceutically important microbes. Sterilization methods in fermentation techniques, air, gas, culture medium sterilization. Steam-filtration and chemicals. Types and constituents of fermentative culture medium and conditions of fermentations, Antifoaming devices.
III	Recovery and estimation of products of fermentation- Production of ethanol, acetic acid, glycerol, acetone, butanol and citric acid by fermentation. Production of Enzymes- amylase, protease, lipase,Production of pharmaceuticals by fermentation– penicillin, streptomycin, tetracycline, riboflavin, vitamin B12.Beverages-wine, beer and malt beverages
IV	Food Microbiology: Production of dairy products-bread, cheese and yoghurt (preparation and their types).Food borne diseases- Bacterial and Non- Bacterial. Food preservation - Principles–Physical methods: temperature (low, high, canning, drying), irradiation, hydrostatic pressure, high voltage pulse, microwave processing and aseptic packaging, Chemical methods - salt, sugar, organic acids, SO_2, nitrite and nitrates, ethylene oxide, antibiotics and bacteriocins
V	Agricultural Microbiology: General Properties of soil, microorganisms in soil – decomposition of organic matter in soil. Biogeochemical

	cycles, nitrogen fixation, Production of bio fertilizers and its field applications – Rhizobium, azotobacter, blue green algae, mycorrhizae, azospirilium, Production of biofuels (biogas- methane), soil inoculants.
Reading List (Print and Online)	Industrial biotechnology: https://nptel.ac.in/courses/102/105/102105058/ Bioreactors: https://nptel.ac.in/courses/102/106/102106053/ Food Microbiology: https://nptel.ac.in/courses/126/103/126103017/ Agriculture Microbiology: https://www.youtube.com/watch?v=f7UXyVImZ_c
Recommended Texts	• Food Microbiology: An Introduction: 4th edition,**Matthews KR, Kniel KE, Montville TJ;** American Society for Microbiology • Food, Fermentation and Micro-Organisms,2nd edition, **Charles, BW;** Blackwell Science Ltd • Microbiology. 5th edition , **Pelczar MJ, Chan ECS and Krieg NR**; McGraw Hill Book Company. • Text book of Microbiology:11th edition, **Ananthanarayanan R and Paniker CKJ;** Universities Press (India) Pvt. Ltd. • FoodMicrobiology, 3rd edition, **Frazier WC and Westhoff DC;**Tata McGrawHill Publishing Company Ltd, New Delhi • New Methods of Food Preservation:1st edition, **Gould GW**; Springer Manual of Industrial Microbiology and Biotechnology**:** 3rd edition, **Baltz**

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or offbeat situations, Discussion, Debating or Presentations

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	M	M	M	S	M	M
CO 2	S	M	M	S	M	S	L	S	L	M
CO 3	S	S	M	S	M	L	L	S	L	M
CO 4	S	S	M	S	L	M	L	S	M	M
CO 5	S	S	L	S	L	M	L	S	M	M

S-Strong M-Medium L-Low

Course	**LIFC011**
Title of the Course:	**LAB COURSE IN MICROBIOLOGY**
Credits:	**4**
Pre-requisites, if any:	**Practical Skills in Handling Microbial Techniques & Fermentation Process**
Course Objectives	The candidates undertaking this course will learn practical skills in the following areas: 1. Introduction to equipment used in microbiology laboratory 2. Gain practical exposure in aseptic techniques 3. Hands on experience in microbial culture techniques 4. Insights into fermentation technology 5. Enumeration of microbes from soil and air

Course Outcomes	**CO1**.Will competently handle all the equipments used in microbial culture (K3,K4,K5) **CO2**.Gain experience in handling microbial culture hood and related aseptic techniques (K3,K4,K6) **CO3.**Gain hands on experience in handling microbes and know sterilization protocols (K3,K4,K5,K6) **CO4**.Good knowledge about several microbial characterization and staining techniques (K3,K4,K5) **CO5**.Gain experience in methods used in current research such as Western blotting, immunofluorescence & PCR (K3,K4,K5)
Units	
	1. Preparation, sterilization and inoculation methods of media. 2. Maintenance and identification of microbes, biochemical characterization and subculturing techniques. 3. Staining techniques 4. Isolation of microorganisms from air, water and soil samples. 5. Antibiotic sensitivity of microbes, use of antibiotic discs. 6. Microbial production of amylase, protease 7. Microbial production of alcohol and citric acid 8. Production and assay of vitamins –Riboflavin

Reading List (Print and Online)	Klamms Microbiology lab manual https://mospace.umsystem.edu/xmlui/handle/10355/69341
Recommended Texts	Talaro, K., Chess, B., Foundations in Microbiology, 8th Ed. 2. Lammert, John M., Techniques in Microbiology A Student Handbook. Laboratory Manual & Workbook in MICROBIOLOGY, Applications to Patient Care, 10th Edition, Morello, Mizer, Granato, McGraw-Hill, 2008, ISBN: 9780073522531

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or off beat situations, Discussion, Presentations

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**S**	**S**	**S**	**S**	**M**	**M**	**S**	**S**	**S**	**S**
CO 2	**S**	**S**	**S**	**S**	**S**	**M**	**S**	**S**	**S**	**S**
CO 3	**S**	**M**	**S**	**S**	**M**	**L**	**S**	**S**	**L**	**L**
CO 4	**S**	**M**	**S**	**S**	**S**	**M**	**M**	**S**	**M**	**M**
CO 5	**S**	**S**	**S**	**S**	**M**	**M**	**M**	**S**	**M**	**M**

S-Strong M-Medium L-Low

B

Course	LIF C 012
Title of the Course:	**EUKARYOTIC GENE EXPRESSION**
Credits:	**4**
Pre-requisites, if any:	**Knowledge of Basic Cell and Molecular Biology**
Course Objectives	**Course Objectives:** **The major learning objectives of the course are:** 1.To introduce the student to chromosomal organization of genes and non-coding DNA, structural organization, morphology and functional elements of eukaryotic chromosomes, organelle DNA. 2. To identify and give examples of the main classes of DNA binding domains; to know the different classes of eukaryotic promoters and the mechanism by which transcription is initiated; to understand how transcriptional control is achieved through alterations in chromatin structure and epigenetic control, to comprehend the molecular basis of RNAi as a technique for gene silencing, and the role of the noncoding siRNAs and miRNAs in control of gene expression 3. To describe the processing of pre-mRNA, tRNA and rRNA, to discuss how alternative splicing and RNA editing are used to modify gene expression in eukaryotes. 4.To describe factors that influence the lifespan of mRNA and to discuss the implications of mRNA stability in regulation of gene expression. 5.To understand the molecular mechanisms of protein targeting, in particular, to the Endoplasmic reticulum (ER) and the secretory pathway, to explain how a protein or RNA is carried into or out of the nucleus, protein folding and quality control in the ER , protein degradation by ubiquitination .
Course Outcomes	**On successful completion of this course, students should be able to:** **CO1.** Describe the general organization in a typical eukaryotic genome and explain the functional significance of packaging DNA into chromosomes. (K1,K2,K4,K5) **CO2**.Demonstrate a competent grasp of the key concepts of gene expression and understand aspects of the control of eukaryotic gene expression, eukaryotic mRNA synthesis, processing, editing , decay, the

	structure, catalytic mechanisms of ribozymes and their therapeutic potential. (K1,K2,K4,K5) **CO3.** Demonstrate a critical understanding of the formation of the noncoding si RNAs and miRNAs and extend and apply the understanding of RNA interference technique for creating knockouts. (K1,K2,K4,K5) **CO4.** Recognize the inevitable errors that could occur due to the complexity of multiple processes involved in gene expression and the quality control mechanisms that have evolved to ensure fidelity. (K1,K2,K4,K5) **CO5.** Understand the importance of the process of directing each newly made protein to a particular destination in a properly folded form, which is critical to the organization and functioning of eukaryotic cells and has a fundamental impact on human health, since many diseases are associated with transport defects. (K1,K2,K4,K5)
Units	
I	The eukaryotic genome **:**The chromosome structure – Histones, Nucleosome, chromatin organization,chromatin remodeling, DNAase hypersensitive sites, genome organization – the C-value paradox, repetitive sequences, gene amplification, significance of introns, loops, domains and scaffolds in eukaryotic DNA, telomeres, organelle genomes – mitochondrial and chloroplast genome.
II	Eukaryotic Transcription**:** Initiation, promoter elements, RNA polymerases, transcription factors, regulatory sequences in eukaryotic protein – coding genes, CpG islands, enhancers. Regulation of eukaryotic transcription – Response elements, DNA-binding motifs, steroid receptors, association of methylation and demethylation with gene expression, molecular mechanism of eukaryotic transcription control- histone acetylation phosphorylation- epigenetics- noncoding RNA- regulatory RNA, RNA interference mechanisms-siRNA and miRNA
III	RNA processing: mRNA 5’ cap and poly-adenylation, splicing, spliceosome assembly, alternative splicing- sex determination in Drosophila, processing of tRNA and rRNA, self-splicing, ribozymes-hammerhead ribozyme.

IV	Degradation of mRNA, nonsense-mediated decay, Exon-junction complex, signal- mediated transfer of mRNA through nuclear pores, RNA editing- substitution and insertion/deletion editing, regulation of mRNA processing, stability of mRNA.
V	Post-and Co-translational modification of proteins: Comparison of prokaryotic translation with eukaryotic translation, Protein sorting – signal peptides, transport of secretory proteins, Golgi and post-golgi Sorting, coated vesicles, retrograde translocation, targeting of mitochondrial and nuclear proteins, Protein degradation-Ubiquitination of proteins, Protein folding-chaperones
Reading List (Print and Online)	Molecular Biology https://ocw.mit.edu/courses/biology/7-28-molecular-biology-spring-2005/ Molecular Biology https://www.mooc-list.com/tags/molecular-biology https://onlinecourses.swayam2.ac.in/cec20_ma13/preview
Recommended Texts	• Lewin's Genes XII : 12th edition. **Krebs JE, Goldstein ES, Kilpatrick ST ;** Prentice Hall • Molecular Biology of the Gene : 6th edition, **Watson JD , Baker TA, Bell S, Gann A, Levine M, Losick R;** Cold Spring Harbor Laboratory Press • Essential Cell Biology : 3rd edition, **Alberts B, Bray D, Hopkin K, Johnson A, Lewis J, Raff M, Roberts K, Walter P ;** Garland Science • Molecular Cell Biology :**Lodish H, Arnold Berk :8th edition;** W.H.Freeman & Co Ltd

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	S	M	S	M	M	S	S	M
CO 2	S	M	S	M	S	M	M	S	S	M
CO 3	S	S	S	M	S	M	S	S	S	S
CO 4	S	S	S	M	S	M	S	S	S	S
CO 5	S	M	S	M	S	M	M	S	S	M

S-Strong M-Medium L-Low

Course	**LIF E005**
Title of the Course:	**BIOCHEMISTRY OF CELL SIGNALLING**
Credits:	**3**
Pre-requisites, if any:	**Basic Knowledge in Biochemistry**
Course Objectives	The main objectives of this course are : 1. To know the origin and type of cellular signal transduction. 2. To study the significance of protein kinases and protein phosphatases in the regulation of various biological processes. 3. Understand signal transduction mechanisms in various disease conditions. 4. Understand Nuclear Receptor Signalling.
Course Outcomes	**CO1**.Students will understand the role of signaling components in cell signaling process. (K1 , K2) **CO2**.Students will understand the coordination of signal transduction processes in the regulation of cellular homeostasis. (K3 , K5) **CO3**.Apply their knowledge to studying the mechanism of gene regulation (K2 , K3) **CO4**.Students will understand the key signal transduction molecules in a physiological setting and in disease progression. (K1 , K4) **CO5.**The student would be able to extend his/her understanding to drug development. (K6)
Units	
I	Structure and functions of intercellular and intracellular signaling pathways: Steps of intercellular signaling. Regulation of intercellular signaling. Role of hormones in intercellular signaling. Classification of intercellular signaling-Endocrine, paracrine and autocrine Signaling. Molecular tools for intracellular signaling-Receptor, Signaling enzymes, Scaffolding proteins and Second messengers. Regulatory mechanisms of intracellular signaling. Convergence, Divergence, and Cross-talk among Signaling Pathways.
II	Protein kinase and protein phosphatase: Classification, Structure and Regulation of protein kinases. Structure and Regulation of PKA,

	PKB, PKC, Ca2+/Calmodulin- dependent protein kinases, Structure and Regulation of Serine/threonine phosphatase, Protein phosphatase 1 (PP1), Protein phosphatase 2A (PP2A), and Protein phosphatase 2B (PP2B).
III	G - protein coupled signal transduction pathways: Transmembrane Receptors - Structure, Major classes of trimeric G proteins based on Gs unit, mechanism of signal transmission, toxins as tools in characterization of G - protein, GTPase switches, G protein that regulate ion channels; G - protein and gene control.
IV	Signaling and Gene control; TGF receptor; Cytokine receptors and JAK - STAT; Receptor Tyrosine Kinases (RTK), activation of ras, genetic analysis - drosophila eye development; MAPK; Phosphoinositide cascade, NF - kB; signal induced protein cleavage, Down modulation of receptor signaling.
V	Nuclear receptors, Principle of signaling by nuclear receptors, Classification and structure of nuclear receptors, Mechanism of transcriptional regulation by nuclear receptors, transactivation. Steroid hormone signaling- estrogen receptor-genomic and non-genomic actions, SERM.
Reading List (Print and Online)	Signal transduction pathways-Introduction: https://www.youtube.com/watch?v=nKL40I9vFUY Cell signaling: https://nptel.ac.in/content/storage2/courses/102103012/pdf/mod5.pdf G-protein signaling pathways: https://www.youtube.com/watch?v=0OUn9188fM8 Receptor Tyrosine Kinases (RTK): https://www.youtube.com/watch?v=6tRaM4WjJkA
Recommended Texts	• Biochemistry ofSignal Transduction and Regulation **:**Fifth edition ,**Gerhard Krauss;** Wiley • Lewin's Genes XII :12th edition. **Krebs JE, Goldstein ES, Kilpatrick ST;** Prentice Hall • Molecular Biology of the Gene : 6th edition, **Watson JD , Baker TA, Bell S,Gann A, Levine M, Losick R;** Cold Spring Harbor Laboratory Press • Essential Cell Biology : 3rd edition, **Alberts B, Bray D, Hopkin K, Johnson A,Lewis J, Raff M, Roberts K, Walter P** ; Garland Science

	• Molecular Cell Biology **:** 8th edition **,Lodish H, Arnold Berk**; W.H.Freeman& Co Ltd

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or offbeat situations, Discussion, Debating or Presentations

Mapping with Programme Outcomes:

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	**S**	**M**	**S**	**S**	**M**	**M**	**M**	**S**	**M**	**M**
CO 2	**S**	**S**	**M**	**S**	**S**	**L**	**L**	**S**	**M**	**S**
CO 3	**S**	**S**	**M**	**S**	**M**	**M**	**L**	**S**	**M**	**M**
CO 4	**S**	**M**	**M**	**S**	**S**	**L**	**L**	**S**	**M**	**S**
CO 5	**S**	**S**	**M**	**S**	**M**	**M**	**L**	**S**	**M**	**S**

S-Strong M-Medium L-Low

Course	**LIF E 006**
Title of the Course:	**DEVELOPMENTAL BIOLOGY**
Credits:	**3**
Pre-requisites, if any:	**Comprehensive Knowledge of Cell Biology**
Course Objectives	The candidates undertaking this course will understand the concepts of developmental biology. 1. To understand the background of developmental biology 2. To gain in-depth knowledge of various model organisms 3. To gain insight into aspects of stem cell technology 4. To gain insights into morphogenesis and oranogenesis 5. To acquire in-depth understanding of cell death mechanisms and cell fate decision
Course Outcomes	**CO1**.Grasp knowledge about the background of developmental biology (K1,K2,K3) **CO2**.Gain abundant knowledge about model oraganisms and gametogenesis (K1,K2,K5) **CO3**.Gain knowledge about stem cells and their applications in regenerative therapy (K1,K2,K3) **CO4**.Good knowledge about organogenesis (K1,K2) **CO5**.Learn the basics of cell death mechanisms and cell fate decision (K1,K2,K3)
	Units
I	Overview of Developmental biology: Background of Developmental biology - Principles of developmental biology –Potency, commitment, specification, induction, competence, determination and differentiation; morphogenetic gradients; cell fate and cell lineages; stem cells; genomic equivalence and the cytoplasmic determinants; imprinting; mutants and transgenics in analysis of development.
II	Model organisms Gametogenesis – production of gametes, Formation of zygote, fertilization and early development: molecules in sperm-egg

	recognition in animals; embryo sac development and double fertilization in plants;cleavage, blastula formation, embryonic fields, gastrulation and formation of germ layers in animals; embryogenesis, establishment of symmetry in plants; seed formation and germination. Drosophila Developmental biology – Axis formation, Genes & mutation. *C.elegans* – Vulva formation, Axis formation.
III	Regeneration Developmental Biology Stem cells – Definition, Classification, Embryonic and adult stem cells, properties, identification, Culture of stem cells, Differentiation and dedifferentiation, Stem cellmarkers, techniques and their applications in modern clinical sciences. Three- dimensional culture and transplantation of engineered cells. Tissue engineering - skin, bone and neuronal tissues.
IV	Morphogenesis & Organogenesis:Cell aggregation and differentiation in Dictyostelium; axes and pattern formation in Drosophila, amphibia and chick; organogenesis – vulva formation in Caenorhabditis elegans, eye lens formation, limb development and regeneration in vertebrates; differentiation of neurons, post embryonic development- larval formation, metamorphosis; environmental regulation of normal development; sex determination.
V	Cellular senescence and Cell fate decision Cellular senescence – concepts & Frizzled receptor in Development and disease. Diabetes and developmental biology, Cell death pathways in development~~s~~. Markers of important diseases.
Reading List (Print and Online)	Developmental Biology – Gilbert Scott http://bgc.org.in/pdf/study-material/developmental-biology-7th-ed-sf-gilbert.pdf
Recommended Texts	Developmental biology: VIII edition, **Gilbert, S**F; Sinauer Associates, Inc

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	S	M	L	S	S	M
CO 2	M	M	M	M	M	S	M	S	M	M
CO 3	M	M	L	M	M	S	L	S	L	L
CO 4	S	M	L	S	S	M	S	S	M	M
CO 5	S	S	M	S	L	M	M	S	M	M

S-Strong M-Medium L-Low

Course	**LIF C013**
Title of the Course:	**RECOMBINANT DNA TECHNOLOGY**
Credits:	**4**
Pre-requisites, if any:	**Basic knowledge of DNA , Vectors and Restriction Enzymes**
Course Objectives	1. To explain to the students about the basic techniques used in creating recombinant DNA , selection and screening methods. 2. To explain the cloning vectors and cloning strategies. 3. To explain the strategy used in transfer of genes into animal cells, transgenic and genome editing techniques 4. To describe Agrobacterium- and virus- mediated gene transfer and development of transgenic plant technology 5. To describe the applications of recombinant DNA technology – gene therapy, antisense therapy and gene directed enzyme prodrug therapy
Course Outcomes	**CO1**.Know the basic techniques used in creating recombinant DNA , selection and screening methods (K1, K2 & K3) **CO2**. Understand the cloning strategies and cloning vectors (K1, K2 & K3) **CO3**.Understand the strategy used in transfer of genes into animal cells, transgenic and genome editing techniques (K1, K2 & K3) **CO4**.Know about Agrobacterium and virus- mediated gene transfer and development of transgenic plant technology (K1, K2 & K3) **CO5**.Understand the importance and applications of recombinant DNA technology – gene therapy, antisense therapy and gene directed enzyme prodrug therapy (K1, K2 & K3)
Units	
I	Basic techniques: Cutting DNA molecules, Restriction digestion, isoschizomers, joining DNA molecules – DNA ligase, double linkers, adaptors, homopolymer tailing, selection of recombinants and screening – genetic methods, immuno chemical methods, South-Western screening, Nucleic acid hybridization methods, synthesis of probes, radio-active and non-radioactive labelling of probes, (Check techniques paper-repetition) Next Generation Sequencing, *in silico*

	sequence analysis, pulsed field gel electrophoresis.
II	Cloning strategies**:** Cloning vectors – plasmids (pBR 322, pUC 18), phage and M 13, cosmids, phasmids, expression vectors, yeast vectors – YEP, YIP, YRP, YCP and YAC, Genomic DNA libraries, cDNA cloning, cDNA library. Inverse PCR, Hot start PCR, RACE, RAPD. Site directed mutagenesis of cloned genes.
III	Introduction of genes into animal cells**:** Reporter genes, selectable markers, viral vectors – SV 40, Retroviruses and Baculovirus, Adenoviruses, Transferring genes into animal cells in culture, oocytes, eggs, embryos and specific tissues, Gene knockout- transgenics and Genome editing.
IV	Agrobacterium and viruses mediated gene transfer. Ti plasmids-Co integrated vector, binary vector) and RI vector. Types of gene transfer methods in plants- Physical and Chemical methods. Transgenic plant technology – for pest resistance, herbicide tolerance, stress tolerance, delay of fruit ripening and produce commercially important proteins, resistance to fungi and bacteria.
V	Applications of recombinant DNA technology – gene therapy- in vivo, ex vivo and in situ gene therapy, vectors used in gene therapy, production of safe viral vectors for gene therapy, examples- gene therapy for SCID, successes, limitations and failures, antisense therapy, gene-directed enzyme prodrug therapy, production of insulin in *E. coli*.
Reading List (Print and Online)	Recombinant DNA https://ocw.mit.edu/courses/biology/7-01sc-fundamentals-of-biology-fall-2011/recombinant-dna/ Biotechnology: Recombinant DNA and Gene Amplification https://www.lecturio.com/magazine/recombinant-dna-and-gene-amplification/ Gene Therapy-Nature https://www.nature.com › gene therapy
Recommended Texts	• Gene Cloning and DNA Analysis**: Brown TA;** Blackwell Publishing; • Principles of Gene Manipulation and Genomics:**Twyman RM, Primrose SB;** Blackwell Publishing; • An Introduction to Genetic Engineering**: Old RW, Primrose SB;** Blackwell Science • Molecular Cloning – A Laboratory Manual :**Sambrook J, Fritsch, EF , Maniatis T**; Cold Spring Harbor Lab Press.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Solve problems, Observe, Explain

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	**S**	**M**	**M**	**S**	**S**	**M**	**S**	**S**	**S**	**M**
CO 2	**S**	**S**	**M**	**S**	**M**	**S**	**S**	**S**	**M**	**M**
CO 3	**S**	**S**	**L**	**M**	**M**	**S**	**S**	**S**	**M**	**L**
CO 4	**S**	**S**	**L**	**M**	**S**	**M**	**S**	**S**	**L**	**M**
CO 5	**S**	**S**	**M**	**S**	**L**	**M**	**M**	**S**	**M**	**M**

S-Strong M-Medium L-Low

Course	LIF C014
Title of the Course:	**Project and Viva Voce**
Credits:	**8**
Pre-requisites, if any:	**Hands-on Experience in Biochemical Techniques and Basic Molecular Biology Techniques.**
Course Objectives	In the fourth semester, the students will spend most of their time on a research project that would result in a dissertation, an oral presentation and a viva- voce. The project will be done in their chosen subject area, within the department, or in another department/institute carrying out related research. 1.To train the student to review relevant scientific literature and choose a research topic , design his/her own experiments, and to develop competence in research methods in biochemistry; 2. To provide guidance on implementing Good Laboratory Practice (GLP), good work ethics 3. To cultivate the ability to work independently and also as part of a research team. 4.To develop science communication, speaking and writing skills. 5. To maintain accurate and up-to-date records of experimental procedures and results and develop the skills of critical interpretation of experimental data.
Reading List (Print and Online)	What is GLP? https://www.certara.com/knowledge-base/what-is-glp-good-laboratory-practice/ A code of ethics for the life sciences https://pubmed.ncbi.nlm.nih.gov/17703607/

Method of Evaluation:

Dissertation	Viva-voce	Total	Grade
75	25	100	

Methods of assessment:

Application (K3) - Solve problems, Observe, Explain

Analyse (K4)

Evaluate (K5) - Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	S	S	S	S	M	S	S	S	S
CO 2	S	L	M	L	S	S	S	S	M	M
CO 3	S	M	M	M	S	M	L	S	M	M
CO 4	S	S	S	S	S	S	S	S	S	S
CO 5	S	S	S	S	S	S	S	S	S	S

S-Strong M-Medium L-Low

Course	LIF E 007
Title of the Course:	**BASICS OF GENE THERAPY**
Credits:	**3**
Pre-requisites, if any:	**Knowledge about Cell and Molecular Biology and Genetic Disorders**
Course Objectives	Course Objectives: The major learning objectives of the course are: 1.To build a fundamental understanding of gene therapy and the reasons to use gene therapy in different diseases with emphasis on the ethical issues surrounding gene therapy. To describe the characteristics of a disease that is a good candidate for gene therapy 2.To provide knowledge about the types of gene therapy and the pros , cons and ethics of germ line gene therapy 3. To illustrate the use of safe viral vectors which have emerged as effective delivery vehicles for clinical gene therapy. To compare the viral vectors with non-viral vectors and suggest the characteristics of an ideal vector. 4. To demonstrate how gene therapy could be used to cure human disease, taking ADA-SCID, X-linked SCID, Familial hypercholesterolemia and suicide gene therapy for cancer. 5. To highlight the successes and setbacks of clinical gene therapy.
Course Outcomes	**On successful completion of this course, students should be able to:** **CO1**. Understand the relevance of gene therapy, compare the different types of gene therapy, be aware of the ethics of gene therapy, and know the inherited and acquired diseases that could benefit from gene therapy. .(K1,K2,K5,K6) **CO2**. Think critically about the various vectors, the advantages and challenges of using viral and non-viral vectors for clinical gene therapy, and why specific viruses can be chosen to deliver genes to target cells. .(K1,K2,K5) **CO3**. understand the basic technologies employed for gene therapy along with approaches and pitfalls to translating these therapies into clinical

	applications .(K1,K2,K4) **CO4**.Knowlege of the regulatory and commercial aspects of this emerging area. (K1,K2,K4,K5,K6) **CO5.** Assess the risk/benefit of gene therapy for individuals and justify reasons for continuing gene therapy trials despite setbacks (K1,K2,K5,K6)
Units	
I	Introduction- What is gene therapy, why gene therapy, Ethical issues in gene therapy, Prospects of gene therapy.
II	Basic process, types of gene therapy, somatic cell gene therapy - *in vitro*, *in vivo*, *in situ* gene therapy.
III	Viral vectors used in gene therapy – Retroviruses, Adenovirus, Adenoassociated virus General advantages and disadvantages of viral vectors. Production of safe viral vectors.
IV	Non-viral vectors used in gene therapy - Liposomes – cationic and anionic liposomes. Advantages and drawbacks of Non- viral vectors, Transposon mediatedvectors,Antisense therapy.
V	Applications of Gene therapy for ADA-SCID, X-Linked SCID, Familial Hyper cholesterolemia, Suicide gene therapy for cancer, Case studies.
Reading List (Print and Online)	Gene Therapy-Nature https://www.nature.com › gene therapy Gene Therapy Basics https://patienteducation.asgct.org/gene-therapy-101/gene-therapy-basics Advances in cell and gene therapy https://onlinelibrary.wiley.com/journal/25738461 What is gene therapy-FDA https://www.fda.gov/vaccines-blood-biologics/cellular-gene-therapy-products/what-gene-therapy

Recommended Texts	
Recommended Texts	• Gene Cloning and DNA Analysis :6th edition**, Brown TA** ; Wiley. • Principles of Gene Manipulation and Genomics **:** 7th edition, **Primrose SB, Twyman RM;** Blackwell Publishing . An Introduction to Genetic Engineering : 7th edition, **Primrose SB** ; Wiley- Blackwell.

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:

Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Create (K6) - Check knowledge in specific or offbeat situations, Discussion, Debating or Presentations

Mapping with Programme Outcomes:

	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**
CO 1	**S**	**S**	**S**	**L**	**S**	**M**	**S**	**S**	**S**	**S**
CO 2	**S**	**S**	**S**	**M**	**S**	**M**	**S**	**S**	**S**	**S**
CO 3	**S**	**S**	**S**	**M**	**S**	**M**	**S**	**S**	**S**	**S**
CO 4	**S**	**S**	**S**	**L**	**S**	**M**	**S**	**S**	**S**	**S**
CO 5	**S**	**S**	**S**	**L**	**S**	**M**	**S**	**S**	**S**	**S**

S-Strong M-Medium L-Low

Course	LIF E 008
Title of the Course:	**MOLECULAR BASIS OF DISEASES AND THERAPEUTIC STRATEGIES**
Credits:	**3**
Pre-requisites, if any:	**Knowledge of Human Physiology, Metabolism and Clinical Biochemistry**
Course Objectives	1. To understand the concepts of the mechanisms involved in regulation of blood sugar and management of diabetes mellitus 2. To gain in-depth knowledge of the mechanisms of cancer and of tumor metastasis 3. The student will review the basic organization of the central and peripheral nervous system that coordinate the sensory and motor functions of the body. In addition, the student will explore impaired features underlying the major neuropathological complications. 4. To gain knowledge in renal diseases 5. To understand the mechanisms involved in cardiac disorders
Course Outcomes	**On completion of this course the student will be able to understand** **CO1.**Overall view about the complications of diabetes mellitus and its management. **CO2.**Comprehensive understanding of the concepts of cancer biology and implicating the theoretical concepts for further research **CO3**.Understand and appreciate the pathophysiology of conditions affecting the nervous system. **CO4**.A thorough knowledge of renal and cardiac diseases with emphasis related to mechanistic aspects and therapeutic interventions. **CO5.** A thorough knowledge on the experimental models of non-communicable diseases that will be applied for future research or project dissertation. An in-depth knowledge on development of drugs against non-communicable diseases.

Units	
I	Mechanism of blood sugar regulation in human body. Pathophysiology of Type I and II diabetes, Diabetes – investigation methods for the diagnosis of diabetes. Nutritional care. Complications related to diabetes – Diabetic cardiovascular disease, retinopathy, neuropathy and nephropathy. Cellular and molecular mechanism of development of diabetes- Management of Type I and Type II diabetes, drugs for the treatment of diabetes.(K1,K2,K3,K4,K5)
II	Biology of cancer: Overview of hallmarks of cancer. Tumorigenesis, Tumor progression and mechanism of Metastasis. Proto-oncogene to oncogene. Oncogene- myc and src family. Tumor suppressor gene-Rb and p53 pathway in cancer. Diagnosis- Non-invasive imaging techniques, Tumor diagnosis, Interventional radiology, New imaging technique, Molecular techniques in cancer diagnosis.- treatment of cancer- surgery, radiotherapy, chemotherapy, hormonal treatment, and biological therapy. Introduction to personalized medicine. .(K1,K2,K3,K4,K5)
III	Brain- neuronal network- memory- Neurogenerative diseases- Parkinson and Alzheimer Disease- molecular understanding of the neurodegenerative diseases- treatment modalities. .(K1,K2,K3,K4,K5)
IV	Acute and chronic renal failure, glomerular diseases–glomerulonephritis, nephritic syndrome, diabetes insipidus, diagnosis of kidney disease.(K1,K2,K3,K4,K5)
V	Introduction to cardiovascular diseases, Lipids and lipoproteins in coronary heart disease-cardiac enzymes, Molecular changes during cardiac remodeling – hypertrophy of hearts – heart failure- treatment modalities.(K1,K2,K3,K4,K5)
Reading List (Print and Online)	The Biochemical basis of disease:2018,**Barr AJ;** Portland Press Biochemical Basis of Diseases https://www.biologydiscussion.com/diseases-2/biochemical-basis-of-diseases/44276

Recommended Texts	Wills' Biochemical Basis of Medicine: 2nd edition, Thomas H, Gillham B;Elsevier Molecular Biochemistry of Human Diseases,2021, Feuer G ,de la Iglesia F; CRC Press

Method of Evaluation:

Sessional I	Sessional II	End Semester Examination	Total	Grade
20	20	60	100	

Methods of assessment:
Recall (K1) - Simple definitions, MCQ, Recall steps, Concept definitions

Understand/ Comprehend (K2) - MCQ, True/False, Short essays, Concept explanations, Short summary or overview

Application (K3) - Suggest idea/concept with examples, Suggest formulae, Solve problems, Observe, Explain

Analyse (K4) - Problem-solving questions, Finish a procedure in many steps, Differentiate between various ideas, Map knowledge

Evaluate (K5) - Longer essay/ Evaluation essay, Critique or justify with pros and cons

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	S	S	M	M	S	S	S	S	S
CO 2	S	M	S	L	M	M	M	M	M	S
CO 3	S	S	M	L	S	S	M	M	S	M
CO 4	S	M	M	M	M	M	S	S	M	S
CO 5	S	S	M	M	S	M	M	M	S	S

S-Strong M-Medium L-Low

*** **The syllabus was prepared by all faculty of the Department of Biochemistry**